

Problem

A balanced wye-connected load is connected to the generator by a balanced transmission line with an impedance of $0.5 + j2 \Omega$ per phase. If the load is rated at 450 kW, 0.708 power factor lagging, 440-V line voltage, find the line voltage at the generator.

Step-by-step solution

Step 1 of 3

Transmission line impedance $Z = (0.5 + j2) \Omega$

Rating of load 450kW, 0.708 power factor lagging

Write the expression for complex power.

$$S = \sqrt{3} V_L I_L \angle \theta$$

$$I_L = \frac{|S| \angle -\theta}{\sqrt{3} V_L} \dots\dots (1)$$

Write the expression for real power.

$$P = S \cos \theta$$

$$|S| = \frac{P}{\text{pf}} \quad (\text{since pf} = \cos \theta)$$

Substitute 450 kW for P and 0.708 for pf .

$$|S| = \frac{450 \times 10^3}{0.708}$$

$$|S| = 635.59 \times 10^3$$

$$|S| = 635.59 \text{ kVA}$$

Step 2 of 3

Substitute 635.59 kVA for $|S|$ and 440 V for V_L .

$$I_L = \frac{(635.59 \times 10^3) \angle -(\cos^{-1}(0.708))}{\sqrt{3}(440)}$$

$$I_L = (833.99) \angle -45^\circ$$

$$I_L = 834 \angle -45^\circ \text{ A}$$

Step 3 of 3

Write the expression for Line voltage at the generator.

$$V_{L(\text{gen})} = V_L + I_L (Z)$$

$$\begin{aligned} V_{L(\text{gen})} &= 440\angle 0^\circ + (834\angle -45^\circ)(0.5 + j2) \\ &= 440\angle 0^\circ + 1719.335\angle 30.96^\circ \\ &= 2108.829\angle 24.798^\circ \\ &= 2.109\angle 24.8^\circ \text{ kV} \end{aligned}$$

Therefore, the line voltage at the generator is $2.109\angle 24.8^\circ \text{ kV}$.